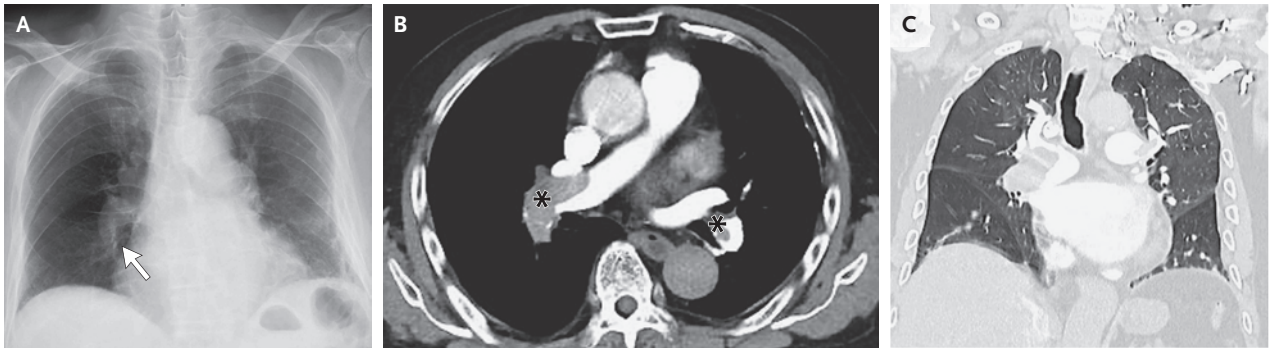
IMAGES IN CLINICAL MEDICINE

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# Westermark's Sign and Palla's Sign in Pulmonary Embolism




A 90-year-old man with atrial fibrillation and dementia presented to the hospital with sudden onset of dyspnea that had begun 1 hour earlier. Before admission, he had not been taking anticoagulation therapy on the basis of discussions of his preferences with his primary care physician. Physical examination was notable for tachypnea and tachycardia. On chest radiography, the pulmonary vasculature was not visible in the right lung fields (Panel A) — a finding known as Westermark's sign, which indicates an area of oligemia distal to a pulmonary embolism. In addition, the right descending pulmonary artery was enlarged (Panel A, arrow) — a finding known as Palla's sign, another rare finding associated with pulmonary embolism. An electrocardiogram showed T-wave inversions in the inferior leads, and the D-dimer level was elevated. Computed tomographic (CT) pulmonary angiography of the chest confirmed the presence of pulmonary emboli in both pulmonary arteries, predominantly on the right side (Panel B, asterisks). The CT scan also showed oligemia in the lung parenchyma on the right side (Panel C). A diagnosis of pulmonary emboli was made. Treatment with apixaban was started, and the patient's condition improved quickly. At a 5-month follow-up visit, he was doing well.

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